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Multiproduct Resin Reuse for Clinical and Commercial Manufacturing—Methodology and Acceptance Criteria

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ABSTRACT: Chromatography resins used for purifying biopharmaceuticals are generally dedicated to a single product. In good manufacturing practice (GMP) facilities that manufacture a limited amount of any particular product, this practice can result in the resin being used for a fraction of its useful life. A methodology for extending resin reuse to multiple products is described. With this methodology, resin and column performance, product carryover, and cleaning effectiveness are continually monitored to ensure that product quality is not affected by multiproduct resin reuse (MRR).

1. Resin and column performance is evaluated in terms of (a) system suitability parameters, such as peak-shape and transition, and height equivalent theoretical plate (HETP) data; (b) key operating parameters, such as flow rate, inlet pressure, and pressure drop across the column; and (c) process performance parameters, such as impurity profiles, product quality, and yield. Historical data are used to establish process capability limits (PCLs) for these parameters. Operation within the PCLs provides assurance that column integrity and binding capacity of the resin are not affected by MRR.
2. Product carryover defined as the carryover of the previously processed product (A) into a dose of the subsequently processed product (B) ($CO_{A \rightarrow B}$), should be acceptable from a predictive patient safety standpoint. A methodology for determining $CO_{A \rightarrow B}$ from first principles and setting acceptance limits for cleaning validation is described.
3. Cleaning effectiveness is evaluated by performing a blank elution run after inter-campaign cleaning and prior to product changeover. The acceptance limits for product carryover ($CO_{A \rightarrow B}$) are more stringent for MRR than for single-product resin reuse. Thus, the inter-campaign cleaning process should be robust enough to consistently meet the more stringent acceptance limits for MRR. Additionally, the analytical methods should be sensitive enough to adequately quantify the concentration of the previously processed product (A) and its degradants in the eluent.

General considerations for designing small-scale chromatographic studies for process development are also described. These studies typically include process-cycling runs with multiple products followed by viral clearance studies with a panel of model viruses. Small-scale studies can be used to optimize cleaning parameters, predict resin performance and product quality, and estimate the number of multiproduct purification cycles that can be run without affecting product quality. The proposed methodology is intended to be broadly applicable; however, it is acknowledged that alternative approaches may be more appropriate for specific scenarios.

KEYWORDS: Chromatography, Multiproduct resin reuse, Biopharmaceutical cleaning validation.

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General considerations for designing small-scale chromatographic studies for process development are described. These studies typically include process-cycling runs with multiple products followed by viral clearance studies with a panel of model viruses. Small-scale studies can be used to optimize cleaning parameters, predict resin performance and product quality, and estimate the number of multiproduct purification cycles that can be run without impacting product quality.

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1. Introduction

Recombinant therapeutic proteins, such as monoclonal antibodies (mAb), are made in living organisms, such as Chinese hamster ovary (CHO) cells. The protein of interest is isolated from cellular and media components used in its production. A typical purification process for a mAb is depicted in Figure 1. Depending on the protein being isolated and the types of impurities present in the harvest pool, some of the purification steps in the process may be excluded or substituted with alternative steps that are more appropriate. The purification process typically consists of a few primary isolation steps followed by a final purification step. The primary isolation process begins after the protein of interest is harvested. The resulting product

pool contains the protein of interest as well as cellular components (e.g., host cell proteins and DNA), media components, and viruses that may be present during the production process. In the first purification step, the product pool is run through a Protein A affinity chromatography column. The purpose of this step is to remove media and cellular components and putative viruses. Following this step, the product pool is further purified by ion exchange (IEX) chromatography. This step serves to remove additional contaminants, such as aggregates and DNA. The IEX step is followed by viral filtration and a second IEX step to remove any residual CHO proteins. The batch is then processed through an ultrafiltration/diafiltration (UF/DF) unit to remove small molecules, concentrate the mAb, and exchange the buffer to formulate the purified protein into its final formulation buffer. This is followed by sterile filtration and bulk-fill operations.

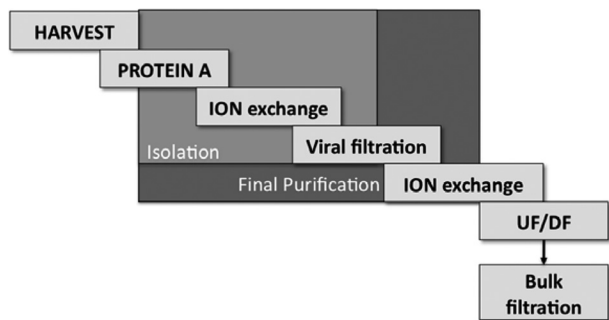


Figure 1

Schematic representation of the purification process for a monoclonal antibody. Protein A affinity chromatography is the first column purification step in the isolation process. This is followed by ion exchange chromatography and viral filtration. The final purification step consists of a second ion exchange column, which is followed by ultrafiltration/diafiltration and sterile filtration.

Chromatography is widely used to purify proteins, as it is efficient, scalable, and reproducible. However, chromatography resins used for purifying biopharmaceuticals are costly and comprise a substantial portion of the raw material costs in biopharmaceutical manufacturing. Further, for clinical and commercial manufacturing, the resins are generally dedicated to a single product. This can result in the resin being substantially underutilized in GMP facilities that make a limited amount of any particular product.

The multiproduct resin reuse (MRR) approach described in this paper can be used to increase resin utilization in all phases of clinical and commercial manufacturing. The increase in resin utilization would typically be substantially greater in clinical manufacturing where fewer lots are manufactured per product. For a clinical facility that manufactures eight new

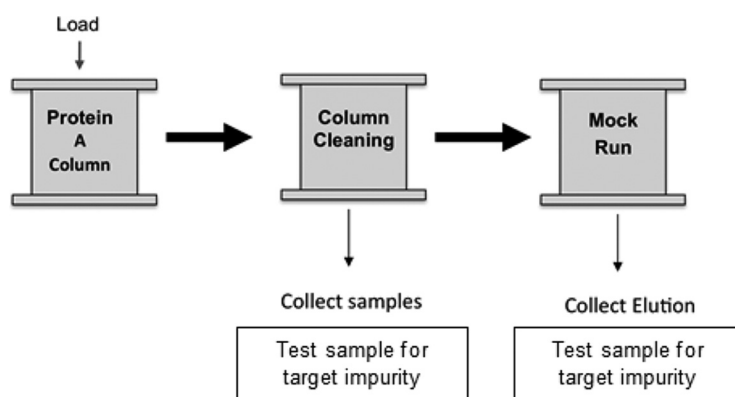


Figure 2

Schematic representation of the steps involved in determining carryover after sample loading and purification on the column, followed by column cleaning and a blank or mock elution run. For the blank elution run, the load cycle is simulated with the equilibration buffer. Samples collected during the elution phase of the blank elution run are tested for the target impurity in the previously manufactured product.

products per year, annualized savings for Protein A resin are estimated to be US\$2,100,000. This estimate is based on \$12,000/L of Protein A resin, a 30 cm column, 25 L per column pack, six cycles per product, and resin reuse for 48 cycles; thus, the cost of the resin per product would be $\$12,000/\text{L} \times 25 \text{ L}$ or \$300,000. It follows that if the resin is re-used for eight products, or 48 cycles per year, instead of being discarded after every product, or 6 cycles, the annualized savings would be \$2,100,000. The savings would be commensurately lower for resins that cost less, such as IEX resin.

A methodology for extending resin reuse to multiple products is described in this paper. With this methodology, resin and column performance, product carryover, and cleaning effectiveness are continually monitored to ensure that product quality is not affected by MRR.

2. Resin, Column, and System Performance

Resin performance is evaluated in terms of the effectiveness of the resin after a given number of lots. The performance of the resin is evaluated relative to established process capability limits (PCLs) for key product quality attributes, such as impurity profiles and yield. The PCLs are established on the basis of historical data from full-scale operations. Operation within the PCLs provides assurance that column integrity and binding capacity of the resin are not impacted by MRR. Column performance is evaluated in

terms of system suitability parameters, such as peak-shape and transition, and HETP data; and key operating parameters, such as flow rate, inlet pressure, and pressure drop across the column. Control limits for these parameters are established through the small-scale column-cycling studies described in Section 2.3. Operation within the control limits provides assurance that the column is operating as intended. System performance refers to the performance of the chromatography unit operation as a whole and includes both resin and column performance.

2.1. Leveraging Full-Scale Data to Verify Stable System Performance—Single-Product Resin Reuse

Most manufacturing operations generate substantial amounts of both continuous and discrete data during downstream unit operations that can be used to verify that flow and bed integrity characteristics are operating within acceptable limits. The continuous data include operating parameters such as flow rates, inlet pressures, and pressure drops across the column. Discrete data include profiles of in-line UV, RI, and/or conductivity from the actual batch processing steps, such as load, washes, elution, strip/regeneration phases, as well as salt or solvent HETP pulses. Review or assessment of the peak shapes or buffer transitions is often part of the manufacturing and quality review of chromatographic unit operations. Blank elution runs described in the Appendix are typically used at some frequency—for example,

at the end of a campaign—to confirm that the cleaning process remains effective.

Manufacturing operations also generate a substantial amount of process- or product-specific in-process testing data that can be used to verify resin and column performance. This includes levels of host-cell impurities, such as DNA or host-cell proteins; step yields; process-related impurities, such as residual antifoam or viral inactivation detergent; and product-specific attributes, such as aggregates, half-antibody or light chain/heavy chain variants, charge isoform distribution, and glycoform distributions. Declining trends in step yield or purity or statistically significant increases in host-cell proteins or other impurities can serve as indicators for declining resin or column performance.

All of these data may be used, in whole or in part, for what may be termed “continuous column performance monitoring” of individual columns or “resin lifetime monitoring” of individual resin lots in a given process. These lifetime monitoring programs have some utility within a short campaign but play a more significant role in longer commercial campaigns to demonstrate that a process is in control, both batch to batch and campaign to campaign. In fact, early trends in these data are usually the reason why columns are taken out of service before they have reached their maximum cycle counts as defined by the small-scale column-cycling studies discussed in Section 2.3.

2.2. Leveraging Full-Scale Data to Verify Stable System Performance—MRR

Conceptually, the approach described in the previous section is equally applicable to resin used across different molecules. The in-line system performance monitoring together with product quality and impurity profile trends can be used to monitor the resin and column performance when the system is used for multiple products. Similarly, key system suitability and operating parameters, such as flow and pressure, peak-shape and transition, and HETP data used to verify stable system performance when the resin is dedicated to a single product (i.e., from campaign to campaign), can also be used to verify system performance when the resin is shared among multiple products. The assessment of peaks and transitions would be product- and buffer-specific, but the monitoring approach would be consistent with the approach described in the previous section for single-product resin reuse.

As an example, consider a Protein-A resin column that is used for a five-batch campaign of mAb-A, then a ten-batch campaign of mAb-B, and then again for a seven-batch campaign of mAb-A. Although mAb-A and mAb-B may have different wash, elute, and regeneration solutions, and therefore different flow and pressure characteristics, peak shapes, and peak areas, the lifetime monitoring criteria would be set specifically for each product. The criteria would additionally include an assessment of the performance of the column during the subsequent seven-batch campaign of mAb-A against the initial five-batch campaign of that product. Well-designed control charts would be a valuable aid in these monitoring assessments. Additionally, statistically significant declines in product impurity profiles and step yields can serve as indicators for monitoring resin performance similar to the conventional approach that is used when the resin is dedicated to a single product.

2.3. Design Considerations for Small-Scale Column-Cycling Studies

Small-scale column-cycling studies with representative target product load streams are used to simulate resin reuse at full scale. The studies are used during process development to establish the maximum number of cycles (N_{MAX}) for which the resin can be reused in manufacturing; the studies are also used to establish limits for process performance qualification (PPQ) runs and routine monitoring during clinical or commercial manufacturing. The process cycling runs are typically followed by small-scale viral clearance studies on the reused resin with a panel of model viruses. The results are compared with those obtained with unused resin to confirm that N_{MAX} is acceptable from a viral clearance standpoint.

Although column-cycling studies have traditionally been used for single-product resin reuse, they can be adapted to include multiple products and to establish N_{MAX} for the application of MRR at full scale. In order to support IND and NDA activities, development data from small-scale studies that demonstrate acceptable predetermined column lifetime monitoring parameters—both within and across runs with different products—can provide strong support for any subsequent MRR at full scale. The following questions should be addressed when designing small-scale studies for MRR applications:

- Which classes of molecules need to be processed, e.g., mAbs, small proteins (peptibodies), or fusion proteins?
- If different classes of molecules need to be processed, which resins will be reused and for which class of molecules?
- Are any of the molecules highly active or potent?
- Does one class of molecules represent a “worst-case” for a particular resin; for example, does one class of molecules tend to leave irreversible aggregates on a particular resin or require an unusually harsh buffer or cleaning step?
- How many batches will be run in each campaign? Based on this information, how many campaigns can possibly be run with the same resin?
- Do any cleaning, regeneration, storage, or handling steps need to be simulated at small scale?

The above considerations can be useful in designing small-scale studies that lay the groundwork for full-scale column lifetime monitoring. An example would be a portfolio of several mAbs and a couple of Fc fusion proteins that are captured using a Protein A resin: one mAb (A) has a relatively low step recovery from the resin, and one Fc fusion protein (B) requires a relatively high concentration of NaOH for cleaning. An experimental design that is representative of worst-case processing conditions might include the mAb with the relatively low step recovery (A), followed by the Fc fusion protein that is relatively difficult to clean (B), followed by a typical mAb (C). The initial cycling design could include every possible switch from one molecule to the other (i.e., A-B-C-A-C-B-A), and the subsequent cycling design could include multiple batches of the same product across a mix of campaigns (e.g., 3xA, 3xB, 5xC, 5xB, 5xA, 3xC). The monitored parameters—including mechanical performance, chromatogram and transition shapes, step yields, and in-process product and residual testing—would provide information on the feasibility of the MRR approach for these molecules and baselines for the full-scale parameters. Note that the small-scale studies may need to be repeated if a new molecule with unusual characteristics or requirements is introduced.

A sufficiently robust design of small-scale studies on a given resin can serve as a one-time “proof-of-con-

cept” for the implementation of MRR for that resin at full scale. When using this approach, it would be advisable to perform a risk assessment each time a new product is to be introduced into the facility. This could be a part of the risk assessment that is performed as part of the technology transfer for the new molecule. The risk assessment should address whether the new product presents significant risk to resin performance. If the risk to resin performance is low, the new product could be introduced into the facility without additional studies, and it could simply be tracked under the column lifetime monitoring; however, if the new product has certain unique characteristics, additional small-scale studies with the new product may be warranted to demonstrate that it is suited for resin reuse. Examples of unique characteristics that may warrant additional small-scale studies include a very different class of molecule or a highly potent or sensitizing molecule; an abnormally high titer load; or a molecule with abnormal aggregation tendencies that could contribute to resin cleanability issues. If the results of the small-scale studies indicate that the new product presents tangible risks for MRR, the resins for that product should be dedicated.

2.4. Implementation at Full Scale

Once the MRR approach has been justified through small-scale cycling studies, it can be implemented at manufacturing scale. The decision to implement the MRR approach for a particular resin or column is generally driven by business considerations, such as resin cost and utilization. For example, it may make sense to implement MRR for resins that are relatively expensive and underutilized, which in turn depends on whether the plant runs short campaigns of many different products or long campaigns of a few products. In the latter case, it may make more sense to dedicate resins, because underutilization of the resin is not a significant issue. The implementation of MRR for a given application would be based on closely monitoring the data discussed in Section 2.1.

The accepted approaches to monitoring and assessing these types of data for column performance over the lifetime of the resin and for campaign trends should be readily adaptable to MRR. Online continuous data, such as flow rates and pressure, would be monitored as usual, although the control limits would clearly be adjusted as applicable to each specific product. For example, mAb-A may have an inlet pressure during load of approximately 22 psig, while for mAb-B, this

inlet pressure may be approximately 18 psig at the same flow rate. Control charts would reflect the appropriate control limits for each product in the sequence of MRR runs. An important assessment of ongoing column performance would be to compare load inlet pressure for mAb-A to the control limits for mAb-A each time the column is used to purify mAb-A.

A similar approach would be followed for trended in-process data at full scale. Consider an example where the process capability limits for host-cell proteins (HCP) in a Protein-A eluate for mAb-A and mAb-B are 4–6 ng/mL and 28–32 ng/mL, respectively. A critical part of the system performance assessment at full scale would be that each time mAb-A or mAb-B campaigns are run, the HCP data from each campaign would be evaluated against the corresponding HCP process capability limits established for that product (i.e., 4–6 ng/mL or 28–32 ng/mL). If the in-process HCP data for each product lie within the established PCLs for that product, it follows that column performance is not affected by any of the intervening campaigns of other products.

Note that some system parameters are independent of the products processed on the column; examples include HETP and asymmetry factor. The control charts for such parameters would have a single set of control limits for all products. These control limits would provide a holistic lifetime view of column performance across multiple products.

3. Cleaning Cycle Development

Another key aspect of implementing resin reuse for multiple products is development of the cleaning cycle for changeover between products. The cleaning process for changeover should be robust enough to consistently meet more stringent acceptance limits for multiproduct reuse. General considerations for designing small- and full-scale studies for cleaning cycle development are discussed in this section. A methodology for setting acceptance limits for product carryover is described in the next section.

3.1. Small-Scale Models for Cleaning Characterization

In addition to predicting resin performance and product quality (Section 2.3), small-scale experimental models are also used to develop and optimize cleaning procedures. Experimental models for evaluating clean-

ing at small scale have been described in the literature (1–5). These models can be used to identify suitable cleaning chemistries (6, 7), estimate cleaning times at full scale (1, 2), optimize cleaning parameters and processes (8, 9), and streamline validation requirements for multiproduct equipment (10, 11). In these models, the rate at which the process residue is removed from the soiled surface (i.e., the mass transfer rate) is measured under simulated cleaning conditions. An important step in the development of these models is to identify and scale down the hardest-to-clean (worst-case) location in the equipment (4). The worst-case location is typically an area with poor circulation, such as an occluded area. If the process residue can be adequately removed from the worst-case location, it follows that it can also be adequately removed from other locations in the equipment. Thus, with this approach, the cleaning times obtained at small scale would be indicative of those at full scale, provided that the worst-case location is appropriately identified and simulated at the small scale. Note that the worst-case location can depend on the specific column and equipment configuration; it is therefore necessary to evaluate every system on a case-by-case basis.

3.2. Scalability of Small-Scale Data

The scalability of small-scale cleaning data depends on the accuracy with which relevant experimental conditions are simulated. Experimental parameters for simulating the worst-case location fall into two distinct categories:

- Parameters that can be readily determined from equipment and process data, such as:
 - equipment configuration, materials of construction, and resin;
 - age and prior exposure of resin;
 - postuse parameters, such as hold time and temperature;
 - cleaning parameters, such as wash time and temperature and concentration of cleaning solution; and
 - storage conditions.
- Parameters that typically need to be determined from first principles, such as:
 - dilution of the process fluid during cleaning (i.e., process residue to cleaning solution ratio);

- velocity of cleaning solution at the worst-case location in the system; and
- protein, excipient, and impurity load per unit of resin surface area.

3.3. Prior Work

Hale et al. (12) had first shown that using appropriate cleaning buffers, the protein carryover or cross-product contamination was dramatically reduced from 250 ppm to below detection levels.

In a more recent study (13), a cleaning process was developed to reduce protein carryover associated with the reuse of Protein A resin for MRR. The cleaning procedure involves multiple pH cycling steps and static holds and can reduce carryover of intact protein into the subsequently purified batch to <1 ppm with no measurable impact on product purity or loss of resin-binding capacity. Other results were as follows: leached Protein A <1 ppm; CHO protein <0.5 ppm; and CHO DNA <1.0 pg/mL. All impurities were comparable to historical data and within acceptable limits for single-resin use.

The pH cycling in the above cleaning procedure involves six alternating cycles between acidic and basic washes. Most proteins tend to break down into smaller fragments at high pH, and the fragments tend to be more soluble at a lower pH. Thus, the pH cycles serve to enhance cleaning through protein fragmentation and solubilization. The buffers used for pH cycling are the elution and regeneration buffers, namely, 0.15 M acetic acid (pH 2.9) and 0.1 N sodium hydroxide (pH 12), respectively. The elution buffer washes bound IgG from the Protein A resin, whereas the regeneration buffer cleaves proteins and nucleic acids, both components of the process residue, into small fragments; it also degrades endotoxins and regenerates the resin. Because these buffers are needed for elution and regeneration, their use for pH cycling is very cost-effective. During the static hold steps, the flow is discontinued for a predetermined amount of time. The addition of static holds in the cleaning procedure provides greater residence time without expending additional buffer. The static holds enhance mass transfer from the resin into the buffer, thereby enhancing diffusion-mediated cleaning.

The above cleaning procedure was successfully implemented for MRR in a pilot plant for manufacturing

material for preclinical toxicology studies (13). Other points to consider in developing cleaning cycles for MRR are discussed in the following sections.

3.4. Other Points to Consider

3.4.1. Support Backbone: Resin-cleaning procedures are typically designed to ensure removal of residual material based upon the ligand chemistry and binding mode—for example, use of high salt regeneration for IEX ligands or water for injection (WFI) stripping in hydrophobic interaction chromatography. Cleaning procedures that expose the resin to high pH may result in removal and degradation of nonspecifically bound species on the resin backbone. For MRR applications, further assessment of the specific matrix properties may be warranted to ensure effective cleaning and acceptable carryover of the previously processed product and its degradants (Section 4.3).

The backbone matrices that comprise the resin beads vary from vendor to vendor and in base properties. Sepharose (cross-linked agarose) is a commonly used base material and is generally considered low risk for nonspecific binding. Agarose is chemically resistant to a wide range of conditions and cleaning agents, thereby making it relatively easy to clean. Polymer matrices, such as polymethacrylate (PMA) and divinylbenzene (DVB), are also fairly common and have inherent hydrophobic properties, with DVB being more hydrophobic than PMA. This inherent hydrophobic interaction can result in product and HCP binding to the backbone, potentially increasing the risk of carryover. In the case of PMA, for example, the polymer forms a lattice-type matrix, providing inherent “pockets” where hydrophobic molecules or fragments can bind and become difficult to remove.

Controlled pore glass has an inherent negative surface charge due to the presence of silanol groups in the matrix. This makes it more likely for the product and HCP to bind to the matrix, thereby increasing the potential for carryover of impurities. Thus, in Protein A chromatography, controlled pore glass is generally less effective than sepharose in removing positively charged impurities. This phenomenon has been attributed to stronger ionic interactions associated with the inherently higher negative charge on glass. The properties of controlled pore glass can also alter with time as the number of exposed silanol groups changes with use and as the resin itself deteriorates over time due to attrition. Matrices that do not fit into a specific back-

bone material family, such as ceramic hydroxyapatite, also present their own potential challenges in terms of cleaning and regeneration and need to be assessed on a case-by-case basis for matrix limitations; for example, ceramic hydroxyapatite is not stable below pH 5.5.

3.4.2. Resin Properties and Chemical Resistance:

The properties of the resin should be considered when developing a cleaning procedure for MRR. For example, a cleaning procedure for a DVB-based anion exchanger might include an elution buffer and 2 M NaCl regeneration buffer, followed by a 1 M NaOH sanitization and cleaning step. While aggressive steps such as these would typically be effective in cleaning the resin, a protein molecule with a highly hydrophobic sequence could undergo partial NaOH-induced degradation, resulting in hydrophobic fragments of the protein binding to the DVB matrix and carrying over into the subsequently manufactured product following changeover. In such situations, a wash with an organic solvent may be warranted, as it would aid in breaking hydrophobic interactions. The use of a WFI strip or more aggressive degradation conditions, such as extended exposure or higher temperature, could also be evaluated as potential solutions.

The base resin matrix and the potential for specific interaction should also be part of the risk assessment for MRR applications. The degradation profile of the product can be leveraged to identify conditions to ensure breakdown and clearance, while remaining within the bounds of acceptable treatment of the resin in terms of chemical exposure. For the risk assessment, the worst-case scenario—nonspecific product binding to the backbone matrix—should be assumed, and the ability of the proposed cleaning procedure to eliminate this interaction, and thereby minimize carryover, should be evaluated. While the overall cleaning procedure may not change, this assessment provides additional assurance that is important for MRR.

3.4.3. Scale Up of Cleaning Procedures: As discussed in Section 3.1, resin-cleaning procedures are initially developed using bench-scale columns. The effectiveness of the cleaning is then evaluated with larger columns. As the column is scaled up, the distribution of fluid flow rate and velocity across the cross section of the column tends to become increasingly nonuniform. In large columns, typically about 80 cm or more in diameter, this phenomenon can result in areas of relatively low flow rate where the cleaning is

inefficient. Additionally, areas around the column wall at the top of the bed where the adapter snap ring and sealing gaskets and O-rings are located can harbor small gaps and crevices where the resin can be trapped during packing. These areas can also be difficult to clean, especially if the resin is not compatible with strong cleaning agents, such as sodium hydroxide.

Large-diameter columns may also have areas of “flow shadowing,” where the flow is slightly reduced or lagging in comparison to the overall laminar solvent front through the bed. These types of areas are associated with regions above and below the packing nozzles, and in alignment with bed support screws, if present. Such areas are common in large columns and can be identified via dye testing and bed dissection.

The cleaning times obtained at small scale should be augmented with an appropriate scaling factor to account for the above flow distortion effects in large columns (Section 4).

3.5. Clean-Out-of-Place Processes

While the ideal scenario, from a logistics standpoint, would be to leave the column packed, clean it in place, and confirm cleanliness *in situ*, the risks described above for large columns may make this difficult to justify for highly potent molecules where very stringent cleaning limits are required. In such situations, it may be necessary to unpack the column, clean the hardware and resin separately, and then repack the column for reuse with the subsequent molecule. This clean-out-of-place (COP) option can be implemented in a number of ways, depending upon the column hardware and available equipment. For columns where the head-plate can be readily removed, the resin can be cleaned by reslurrying it with the cleaning solution. The cleaning can be performed manually by reslurrying the resin with a paddle. This provides assurance of full resin contact with the solution and eliminates the issues associated with the aforementioned areas around the upper adapter. Once the desired contact time has been achieved, the resin can be allowed to settle, and the cleaning solution can be removed via an external pump. Alternatively, the head can be lowered until sealed, and the solution can be exchanged via flow through the column.

When the resin cannot be resuspended in the column itself, the resin can be unpacked and transferred to a tank or a portable container, such as a carboy, allowed

to settle, and then manually decanted to facilitate exchange with the cleaning solution. The slurry can be mixed with the cleaning solution for the desired time and then transferred back into the column for further processing. When using this approach, the chemical resistance of the resin should be considered to ensure that the integrity of the resin is not compromised by overexposure to the cleaning solution. Once the resin has been adequately exposed to the cleaning solution, it can be exchanged into a compatible packing solution, if necessary, and the column repacked for additional buffer exchange and use.

An intermediate suspension approach can be used for large diameter columns (>1 m) where the column height is sufficient to resuspend the resin within the column at a 50%–70% slurry ratio. With this approach, the column's upper adapter can be raised to the maximum height, and the resin can be resuspended via flow through the nozzles using a packing station in a recirculation loop. The resin can be resuspended in this manner in the cleaning solution. Then the cleaning solution can be exchanged for repack, if required, and the column can be repacked for reuse. It should be noted that some resins are prone to bead breakage with handling and could be damaged if the recirculation is not carefully controlled. Because this approach involves multiple passes of the resin through the recirculation loop, careful consideration must be given to the resin type and potential for fracture of the resin.

The COP approaches described above are designed for high-risk scenarios, such as highly potent molecules and large-diameter columns that have the potential for inadequate cleaning. Prior cleaning processes used for specific hardware and column diameters can be leveraged to develop new cleaning processes. Sampling of the full-scale packed column cleaning using the proposed cleaning process can be used to determine the level of carryover (Section 4), as well as to ascertain if a COP-based approach may be required.

Another important element of cleaning cycle development and performance monitoring are blank elution runs (refer to the Appendix). These runs may be performed either on a schedule, such as after a predetermined number of runs, or at the end of a campaign. Samples are taken from the blank elution run at appropriate time points to determine the amount of carryover (Section 4). The blank elution runs provide another set of data that can be lever-

aged to assess cleaning effectiveness and product carryover for MRR.

A methodology for determining product carryover and setting acceptance limits for cleaning validation is described in the next section.

4. Evaluating Cleaning Effectiveness

As described in the previous section, small-scale cycling studies can be used to develop resin-cleaning cycles. Additionally, appropriate scaling factors can be applied to ensure that the cleaning cycles will be effective at full scale; for example, depending on the complexity and understanding of the full-scale system, the duration of critical cleaning steps could be increased by a cleaning effectiveness scaling factor typically between 1.2 and 2 (20). The effectiveness of the resulting resin-cleaning procedures should be evaluated as part of performance qualification at full scale. A methodology for evaluating cleaning effectiveness at full scale is described in this section.

4.1. Regulatory Expectations

An important regulatory expectation for cleaning performance qualification of multiproduct equipment is that the potential carryover of the previously manufactured active pharmaceutical ingredient (API) and its degradants—hereinafter referred to as the target impurity—into the subsequently manufactured product should be acceptable from a predictive safety standpoint (14). This criterion is evaluated through a safety-based maximum allowable carryover (MAC) assessment for the target impurity (15). A methodology for performing a MAC assessment for a multiproduct equipment train that includes one or more MRR steps is described in the sections that follow. The results are used to evaluate overall cleaning effectiveness and to determine acceptance limits for the target impurity for swab and rinse samples.

4.2 Identification of Target Impurity

Resin-cleaning cycles are often designed to expose the API to extreme pH (<3 and >12) for an extended period of time. Under these conditions, proteins and other biopharmaceuticals are known to degrade and denature and are therefore likely to become pharmacologically inactive (16). Degradation and inactivation of the API during resin cleaning are assessed by

TABLE I
Possible Outcomes of an Inactivation Study

Outcome of Study	Target Impurity Associated with Product A	Acceptance Limit of Target Impurity per Dose of Product B
No detectable change in activity	API ^a (protein)	ADE ^b of Product A
No detectable activity	Inactive protein	650 µg (15)
Partial inactivation	API and inactive protein	Lower of ADE of Product A or 650 µg

^a Active pharmaceutical ingredient; ^b Acceptable daily exposure.

exposing the process residue to simulated cleaning conditions at bench scale (16, 17). The experiments are designed to simulate full-scale cleaning conditions that are the least conducive (worst-case) for inactivation; for example, the lowest possible temperature and concentration of cleaning solution. The results of the bench-scale studies can justifiably be extrapolated to the full-scale cleaning process because the kinetics of degradation and inactivation are not scale-dependent (i.e., they depend on cleaning parameters that are essentially independent of the spatial coordinates of the system, such as the duration of the cleaning and the temperature and concentration of the cleaning solution). The samples are subjected to analytical testing to determine the degree of degradation and inactivation of the API. The degree of degradation (e.g., API fragmentation) can be evaluated by SDS-PAGE (sodium dodecyl sulfate polyacrylamide gel electrophoresis). API inactivation can be evaluated by methods that measure loss of biological activity or function (e.g., binding sites that are functionally intact), such as a bioassay. These methods measure the relative amount of biologically active product; thus, they can be used to evaluate the degree of API inactivation.

Possible outcomes of the inactivation study are summarized in Table I. Note that target impurity is identified from the results of the study: no detectable change in activity implies that the target impurity is the API; conversely, no detectable activity indicates that the target impurity is inactive protein. If the API is partially inactivated, it follows that there are two target impurities: the API and the inactive protein. Thus, in case of partial inactivation, the assessment described in this section should be performed for both active and inactive protein.

In addition to process impurities, such as active and inactive protein, the clearance of contaminants, such

as cleaning agents, bioburden, and endotoxin, should also be evaluated. Acceptance limits for cleaning agents are set based on their acceptable daily exposures (ADE), and for bioburden and endotoxin, they are generally set based on WFI specifications per USP (i.e., 10 CFU/100 mL and 0.25 EU/mL, respectively).

A safety-based approach for setting acceptance limits for active and inactive protein is described in the next section.

4.3. Acceptable Limit for Target Impurity

The methodology described in this section is based on a commonly used approach for setting acceptance limits for cleaning validation (14, 15). Other approaches have been described in the literature and could be used if applicable. If the target impurity is the API, the acceptable limit of the target impurity per normalized daily dose of the subsequent product (AL_{TI}) is the ADE (also known as permitted daily exposure, PDE) of the API (18); conversely, if the target impurity is the inactive product, AL_{TI} can be set to 650 µg (15):

$$AL_{TI} \text{ (per dose)} = ADE \quad (\text{if the target impurity is the API}) \quad (1a)$$

$$AL_{TI} \text{ (per dose)} = 650 \mu\text{g} \quad (\text{if the target impurity is the inactive product}) \quad (1b)$$

Further, if the API is partially inactivated, the assessment could be simplified by setting AL_{TI} to the lower of the two limits; thus,

$$AL_{TI} \text{ (per dose)} = ADE \quad (\text{if } ADE < 650 \mu\text{g}) \quad (1c)$$

or

$$\begin{aligned} \text{AL}_{\text{TI}} (\text{per dose}) &= 650 \mu\text{g} \\ &(\text{if ADE} > 650 \mu\text{g}) \quad (1\text{d}) \end{aligned}$$

If the ADE of Product A is not available, as in the case of a legacy product, AL_{TI} can be set to 0.1% of the smallest therapeutic dose of Product A (19).

If the protein is not inactivated when exposed to the cleaning conditions, the cleaning parameters can be modified to ensure inactivation of the API. This can be facilitated by running additional small-scale studies to characterize the effect of the cleaning parameters on activity and to determine the cleaning conditions under which the API is inactivated.

4.4. Maximum Allowable Carryover for Target Impurity

The basis for the acceptable limit of the target impurity (AL_{TI}) in Equation 1a through 1d is the dose of the subsequently manufactured product (Product B). In order to determine the MAC of target impurity into a batch of Product B, the basis for AL_{TI} must be changed from dose to batch of the drug product (DP) (i.e., in the final batch of Product B):

$$\begin{aligned} \text{AL}_{\text{TI}} (\text{per batch of DP}) &= \text{AL}_{\text{TI}} \\ &(\text{per dose}) \cdot N \quad (2\text{a}) \end{aligned}$$

where N is the number of doses per batch of the drug product. Further,

$$N = M_{\text{B}}/m_{\text{B}}$$

where M_{B} is the mass of API_{B} in a batch of the drug product, and m_{B} is the mass of API_{B} in a dose of Product B that is administered to a patient in a day. Thus,

$$\begin{aligned} \text{AL}_{\text{TI}} (\text{per batch of DP}) &= \text{AL}_{\text{TI}} \\ &(\text{per dose}) \cdot (M_{\text{B}}/m_{\text{B}}) \quad (2\text{b}) \end{aligned}$$

where $\text{AL}_{\text{TI}} (\text{per dose})$ is defined by Equation 1a through 1d.

Note that if the DP batch is split into smaller batches for filling or other reasons, M_{B} should be the mass of Product B (i.e., API_{B}) in the original batch (i.e., before it is split). Further, if Product B has multiple DP batch sizes, M_{B} should be based on the smallest batch size; if Product B has multiple formulations, m_{B} should be

based on the formulation with the largest therapeutic dose of Product B (i.e., API_{B}). The aforementioned criteria represent worst-case scenarios from the standpoint of exposure to the target impurity and therefore to $\text{AL}_{\text{TI}} (\text{per batch of DP})$. $\text{AL}_{\text{TI}} (\text{per batch of DP})$ in Equation 2b is the MAC of the target impurity into a batch of Product B ($\text{MAC}_{\text{TI} \rightarrow \text{B}}$); thus,

$$\begin{aligned} \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{total}) &= \text{AL}_{\text{TI}} (\text{per dose}) \\ &\cdot (M_{\text{B}}/m_{\text{B}}) \quad (3) \end{aligned}$$

4.5. Maximum Allowable Carryover for Multiproduct Equipment

The total surface area shared between Products A and B has two distinct components: (a) the resin surface area associated with MMR and (b) the surface area associated with multiproduct equipment (MPE) that is reused (e.g., stainless steel and glass). Thus, $\text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{total})$ in Equation 3 can be expressed as follows:

$$\begin{aligned} \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{total}) &= \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{MRR}) \\ &+ \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{MPE}) \quad (4\text{a}) \end{aligned}$$

where $\text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{MRR})$ and $\text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{MPE})$ are components of $\text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{total})$ associated with resin and equipment reuse, respectively, for Products A and B. Equation 4a can be written as:

$$\begin{aligned} \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{MPE}) &= \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{total}) \\ &- \text{MAC}_{\text{TI} \rightarrow \text{B}} (\text{MRR}) \quad (4\text{b}) \end{aligned}$$

The carryover of the target impurity from the reused resin can be evaluated by performing a blank elution run at small scale as described in the Appendix. Samples collected during the elution step of the blank elution run are analyzed for the target impurity associated with Product A. The results are used to estimate the carryover of the target impurity into Product B ($\text{CO}_{\text{TI} \rightarrow \text{B}}$) that can be attributed to MRR:

$$\text{CO}_{\text{TI} \rightarrow \text{B}} (\text{mrr}) = \sum_i (V_{\text{EB}} \cdot C_{\text{EB}}) \quad (5)$$

where V_{EB} is the volume of the elution buffer, C_{EB} is the concentration of the target impurity in the elution buffer, and $\sum_i$ is a summation over all MRR steps associated with the target impurity. The abbreviation mrr indicates that the parameter, in this case $\text{CO}_{\text{TI} \rightarrow \text{B}}$, is based on evaluation at small scale.

$MAC_{TI \rightarrow B}$ (MRR) can be estimated from $CO_{TI \rightarrow B}$ (mrr) by introducing a scaling factor for carryover (SF_{CO}) to account for operational discrepancies between the small-scale and full-scale systems:

$$MAC_{TI \rightarrow B} \text{ (MRR)} = CO_{TI \rightarrow B} \text{ (mrr)} \cdot SF_{CO} \quad (6)$$

where $CO_{TI \rightarrow B}$ (mrr) is the carryover of the target impurity into Product B associated with MRR based on evaluation at small scale. SF_{CO} depends on the complexity and understanding of the full-scale system. Combining Equations 5 through 6 gives:

$$MAC_{TI \rightarrow B} \text{ (MRR)} = \sum_i (V_{EB} \cdot C_{EB}) \cdot SF_{CO} \quad (7)$$

Combining Equations 3, 4b, and 7 gives:

$$MAC_{TI \rightarrow B} \text{ (MPE)} = AL_{TI} \text{ (per dose)} \cdot (M_B/m_B) - \sum_i (V_{EB} \cdot C_{EB}) \cdot SF_{CO} \quad (8)$$

Note that if the carryover is determined by running a blank elution run at full scale, SF_{CO} is equal to 1, and as a result, it can be dropped from Equation 8.

In the next section, Equation 8 is used to determine acceptance limits for swab or rinse samples for cleaning validation.

4.6. Acceptance Limits for Cleaning Validation

$MAC_{TI \rightarrow B}$ (MPE), the MAC for MPE (Equation 8), can be used to determine total organic carbon (TOC) acceptance limits for swab and rinse samples for MPE (e.g., vessels and lines) that is shared between Products A and B.

4.6.1. Acceptance Limit for Swab Samples: The acceptance limit for swab samples (AL_{SWAB}) is determined using the following equation:

$$AL_{SWAB} = [MAC_{TI \rightarrow B} \text{ (MPE)} / SA_T] \cdot (SA_S / V_S) \cdot (SRF \cdot CC) \quad (9)$$

where SA_T is the equipment surface area that is shared between Product A, which is associated with the target impurity, and the subsequently manufactured product (Product B); SA_S is the surface area swabbed (typically 25 cm²); V_S is the volume of the swab extractant solution (typically between 25 and 50 g); SRF is the

fractional swab recovery factor for the target impurity (0–1; dimensionless); and CC is the fractional carbon content of the target impurity (0–1; dimensionless). To obtain AL_{SWAB} in parts per million carbon in the extractant (i.e., microgram carbon per gram extractant), $MAC_{TI \rightarrow B}$ (MPE) and SA_T must be in units of micrograms and centimeter square, respectively.

The most conservative approach for estimating SA_T is to include all shared product-contact equipment in the calculation (e.g., all cell culture, harvest, and purification equipment for drug substance manufacturing); however, with appropriate justification, some or all of the upstream equipment can be excluded from the calculation for estimating SA_T (e.g., equipment upstream of a particular purification step can be excluded if the target impurity is adequately removed by the subsequent purification steps).

4.6.2 Acceptance Limit for Rinse Samples: The acceptance limit for rinse samples (AL_{RINSE}) is determined using the following equation (21):

$$AL_{RINSE} = [MAC_{TI \rightarrow B} \text{ (MPE)} / V_T] \cdot [RRF / (1 - RRF) \cdot CC] \quad (10)$$

where V_T is the cumulative final rinse volume for product contact equipment shared between Product A, which is associated with the target impurity, and the subsequently manufactured product (Product B); RRF is the fractional rinse recovery factor for the target impurity (0–1; dimensionless); and CC is the fractional carbon content of the target impurity (0–1; dimensionless). To obtain AL_{RINSE} in parts per million carbon in the rinse water (i.e., microgram carbon per gram of rinsate), $MAC_{TI \rightarrow B}$ and V_T must be in units of micrograms and grams, respectively.

As with estimating SA_T , the most conservative approach for estimating V_T is to include all shared product-contact equipment in the calculation; however, with appropriate justification, some or all of the upstream equipment can be excluded from the calculation for estimating V_T .

5. Conclusion

The current approach of dedicating chromatography resins to a single product can result in substantial underutilization of the resin. A proposed methodology for extending resin reuse to multiple products was described. With this methodology, resin and column performance,

product carryover, and cleaning effectiveness are continually monitored to ensure that product quality is not impacted by MRR. Control limits for operating parameters can be established through small-scale column-cycling studies. Operation within the control limits provides assurance that the column is operating as intended. Historical data are used to establish PCLs for key performance attributes. Operation within the PCLs provides assurance that column integrity and binding capacity of the resin are not impacted by MRR. In addition to increasing resin utilization, the proposed resin reuse approach would reduce column packing and resin handling and storage requirements.

A methodology for implementing MRR in clinical and commercial manufacturing was described. Scientific criteria for determining product carryover from first principles and setting acceptance limits for cleaning validation were also described. General considerations for designing small-scale chromatographic studies for process development were discussed. These studies typically include process-cycling runs with multiple products followed by viral clearance studies with a panel of model viruses. The small-scale studies are used to optimize cleaning parameters, predict resin performance and product quality, and estimate the number of multiproduct purification cycles that can be run without impacting product quality.

The approaches described in this document are recommendations that can be substituted with alternative approaches that may be more appropriate for specific scenarios.

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Conflict of Interest Declaration

The authors declare that they have no competing interests.

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Appendix

Blank Elution Run for Determining Carryover of Target Impurity from Resin

A blank or mock elution run can be used to collect a representative sample after the resin is cleaned. The sample is analyzed for the target impurity, and the results are used to determine the carryover of the target impurity into Product B for that MRR step. The blank elution run is performed following Product A processing and routine column cleaning (Figure 2). It involves normal processing steps for Product B, with the exception that the load cycle is simulated with elution buffer instead of the Product B load pool. A sample of the elution pool is then collected to determine the concentration of the target impurities. The results are used to estimate the total carryover of the target impurities into Product B. Note that the samples may need to be adjusted to pH 5–5.5 (with 1.5 M Tris base buffer, for example) and then treated with a suitable reagent (such as 0.1% polysorbate, 0.05% sodium azide, or 0.05% phosphoric acid) to pre-

vent a false-negative result owing to surface adhesion of the target impurity. If necessary, the elution pool can be concentrated to facilitate detection of the target impurity.

The blank elution runs may be performed at either bench scale or full scale. In either case, the experimental parameters should be representative of those used to manufacture Product B (e.g., volume and excipient composition of loading and elution buffers and ratio of buffer to resin volume). Note that for the blank elution run, the carryover of the target impurity into the elution buffer represents a worst-case scenario from the standpoint of cross-contamination. This is because the elution buffer does not contain any protein, and therefore, from a mass transfer standpoint (i.e., in terms of the net driving force for molecular diffusion), it is more favorable for the target impurity to elute into the blank elution buffer. Thus, under the same operating conditions, the carryover of the target impurity into the blank elution buffer would be greater than that into the subsequently manufactured product.

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